Model of Foucault Pendulum: Unlocking the Mysteries of Earth's Rotation

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Abstract

In this study the motion of the Foucault pendulum is modeled numerically to demonstrate Earth's rotation. Using numerical methods such as the 4th order Runge-Kutta method, I was able to compute the pendulum's motion but also calculate the precession rate at various latitudes. These latitudes included the equator (0 degrees), Japan (36.2048), New York (40.7128), and Antarctica (82.8628). The precession rates varied with latitudes due to the Coriolis effect which shifts the oscillation plane due to Earth's rotation. The calculated precession rates were compared to theoretical values obtained from Earth's rotational dynamics. The analysis computed a minimal error of 18.3%, showing that the observed and theoretical values were consistent with one another. I also analyzed the pendulum's energy numerically to prove that the total energy remains conserved over time. This provides further evidence of the numerical model and aiding the demonstration of Earth's rotation. All and all, the results confirmed that Earth is indeed a rotating sphere as precession values varied with latitude, debunking the flat Earth theory.

Introduction

The Foucault pendulum was named after French physicist Leon Foucault, who introduced the pendulum in 1851 in the Meridian of the Paris Observatory. This experiment was used to visually demonstrate the Earth's rotation through the steady precession of its oscillation plane that changes with latitude. This experiment was the first to transform physics as a whole by providing a real world observable experiment of theoretical concepts that made it easier for society to understand as a whole. It was also one of the first observable pieces of evidence that could challenge the flat Earth claims. The flat Earth theories don't consider precession which is a crucial component to the Foucant pendulum behavior. The pendulum's trajectory changes as Earth rotates beneath it, resulting from the Coriolis force which is 2*Omega sin (phi), where Omega is Earth's angular velocity and phi is the latitude. The Coriolis force is based on precession and is strongest at the poles but weak at the equator. From this you can assume the pendulum's oscillation plane varies with latitude. Using various latitudes such as New York, Japan, Equator, and Antarctica we were able to prove this statement. At the North/South poles (90 degrees) the precession does a full 360 in a day but at the equator (0 degrees) there is no precession since the Coriolis force is zero. This study used different computational methods in order to model the Foucault pendulum motion/precession at different latitudes. I used the 4th order Runge Kutta method to solve the governing equation motion, calculate precession rates, and prove energy conservation. Energy conservation evaluated numerically showed that energy remained constant over which proved the simulation stability. Comparing the rates to their theoretical values was significant as it led to a deeper understanding of Earth's rotational dynamics but also revealed the role of the oscillatory system in this model. This analysis emphasizes the advantages that come from the Foucault pendulum, such as being able to explore rotational dynamics, understand Earth's rotation, and prove the Earth is indeed a spinning sphere.

Procedure/Methods

The motion of the Foucault pendulum was modeled using the 4th order Runge-Kutta(RK4) method. I chose this method as it's the most accurate when solving ordinary differential equations. It splits the equations into acceleration and velocity terms which produces iterative computation of the pendulum's motion over small time steps. The equations were discretized with a time step of (dt) 0.0003 to ensure numerical stability and minimal numerical error. In order to model the motion of the Foucault pendulum, you first need to set up the governing equation of motion. Dvx and dvy is the acceleration in the x direction and y direction

dvx=2w sin(phi) dydt-(g/L)*x dvy= -2w* sin(phi) dxdt-(g/L)*y

The Coriolis force (2*Omega sin (phi)) is introduced due to Earth's rotation which causes the trajectory changes in the pendulum's plane of oscillation. The equations depend on Earth's angular velocity (w) and latitude (phi). -g/L is the restoring force due to gravity. X and y are the positions of the pendulum in two dimensions. dx/dt and dy/dt are the velocities in the x and y directions. w= 2pi/86400 rad/s which is the earth's angular velocity. G is the gravity being 9.8. L is the length of the pendulum being equal to 64. Next, The RK4 was used to numerically solve the equation of motion. The positions(x,y) and velocities (dx/dt,dy/dt) were all calculated repeatedly at each time step to get the oscillation dynamics. The algorithm splits the calculation into four different steps, k1, k2, k3, k4 for each component of the equation (x,y,vx,vy). It updates the pendulum's position and velocity from what was calculated in the beginning (dvx,dvy). The initial conditions being: x[0] = x0, y[0] = y0, dxdt [0] = vx0, dydt [0] = vy0, and
energy[0]=½ m (dxdt[0]^2+dydt[0]^2)+m g (L- sqrt(L^2-x[0]^2-y[0]^2))
The total mechanical energy of the pendulum was evaluated numerically at each time step during the modeling. It combines both the potential and kinetic energy to ensure conservation is conserved throughout the experiment. The potential energy (mgh) is the mass=10 times gravity times h. Which h is just the height of the pendulum above the equilibrium. We get h to be equal to L minus the square root of L^2-x^2-y^2 using L=64 and pythagorean theorem. Then add that to our kinetic energy being ½ times mass times the pendulum's velocity in the x and y direction. (dxdt^2+dydt^2) The theoretical precession rate (theory) is given from Earth's rotational dynamics (w sin(phi)). sin(90) =1 so the pendulum does a 360 every 24 hours while sin(0)=0 so it does rotate at all just swings in place. On the other hand, the modeled precession is calculated numerically as the displacement of the angular components of the pendulum's oscillation over time (T=100). (arctan (y [-1]/x[-1])/T) You then compare the theoretical and modeled precession rate to compute how accurate the model was (error) for each latitude. (precession-theory)/theory)*100) My error was 18.3%, which is slightly high but manageable after testing various parameters and getting up to an error of 4000%. The model was computed at 4 various latitudes with the trajectory of the pendulum being represented in x/y plots and the energy being proven conserved through energy/time graphs. These methods/equations display the motion of a Foucault pendulum to show proof of Earth's rotation.

Results

The plots for the trajectory of the pendulum at various latitudes showed the steady procession of the oscillation plane. The higher the latitude the more you could see the precession, while at the equator there was none to be observed. The plots aligned perfectly with the behavior expected theoretically. When comparing the modeled precession rates with the theoretical precession I got a minimal error of 18.3%, confirming the accuracy of the numerical models. In some cases I obtained larger errors due to the values entered for various parameters such as length, timestep, and total time. So I had to fiddle with the numerical values to reduce the overall error, producing accurate results. The energy/time graphs for each latitude I analyzed to remain constant over time, pointing to energy conservation. The total mechanical energy was proven using the plotted graphs. Figure 1 shows the detailed comparison of theoretical vs modeled precession rates at all the latitudes. The values are pretty close to one another with some discrepancies at the higher latitudes. This could be a production from limitations in numerical accuracy. The Coriolis force is the main character that causes the precession rates to change with latitude. These results showcase the crucial role the Foucault pendulum plays in order to get a deeper understanding on Earths' rotation dynamic and conservation of energy. The Foucaut pendulum provides enough evidence to debunk flat Earth theories, which strengthens the experiment's reliability and importance for society as whole. Overall, the experiment provided a successful result to prove Earth's Rotation.

Latitude	0	36.2048	40.7128	82.8628
Error	18.3%	18.3%	18.3%	18.3%
Modeled Precession	0	3.5e-05	3.9e-05	5.9e-05
Theory Precession	0	4.3e-05	4.7e-05	7.2e-05

*Figure 1- Presents the results of the Foucault Pendulum simulation at various latitudes

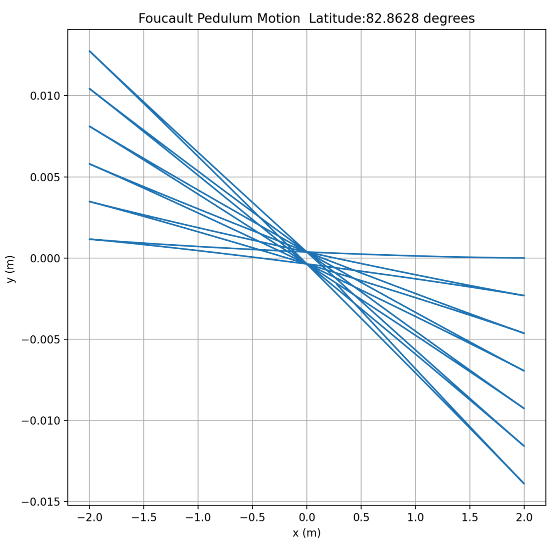


*Figure 2- latitude Antarctica trajectory of pendulum's oscillation strong indication of Coriolis force

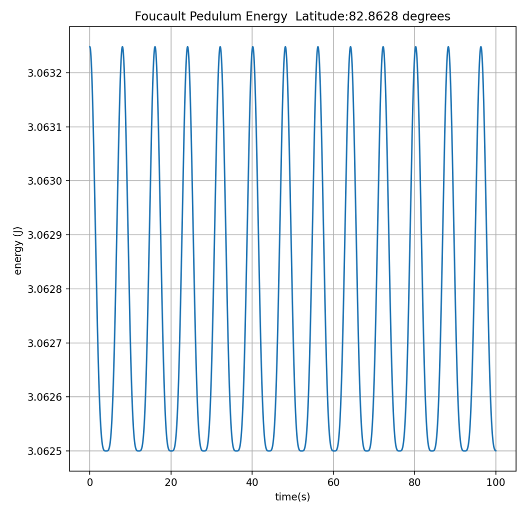


*Figure 3- Antarctica energy is constant as it starts and end at the same location

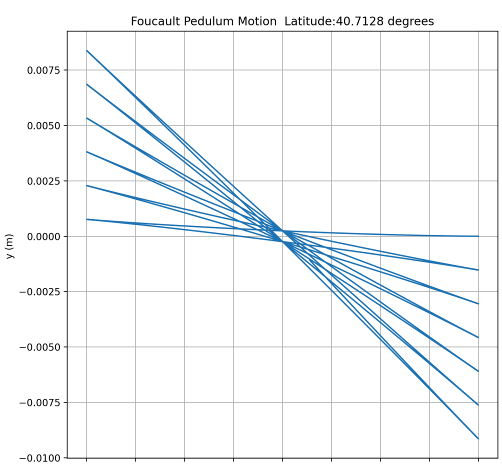
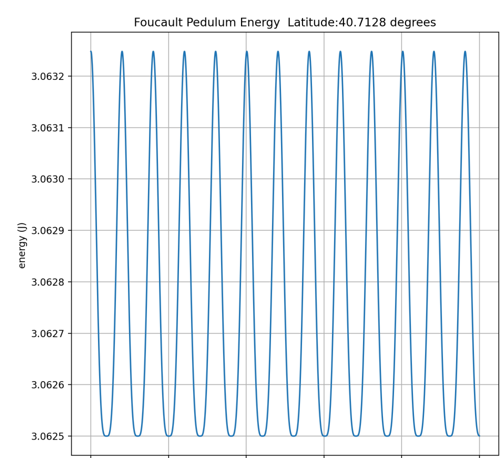


*figure 4- New York trajectory graph show how It affects the earth's rotation on the pendulum' motion.

*figure 5- New York energy proves the conservation of energy throughout the simulation.



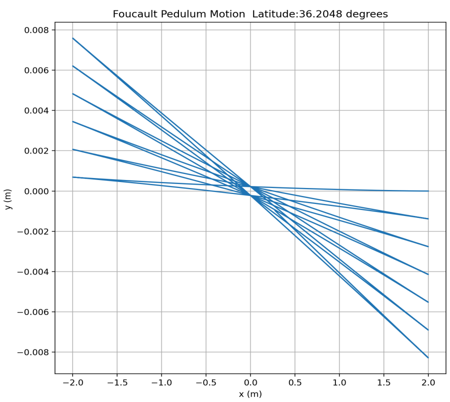


*figure 6- Japan trajectory graph show how It affects the earth's rotation on the pendulum' motion.

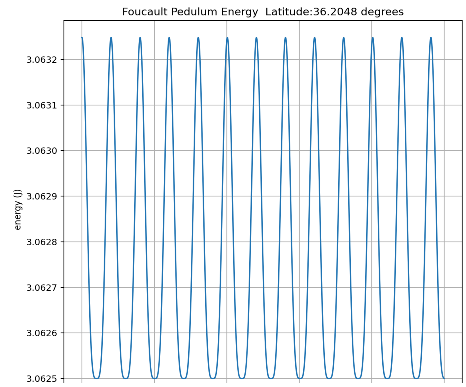


*figure 7- Japan energy proves the conservation of energy throughout the simulation.

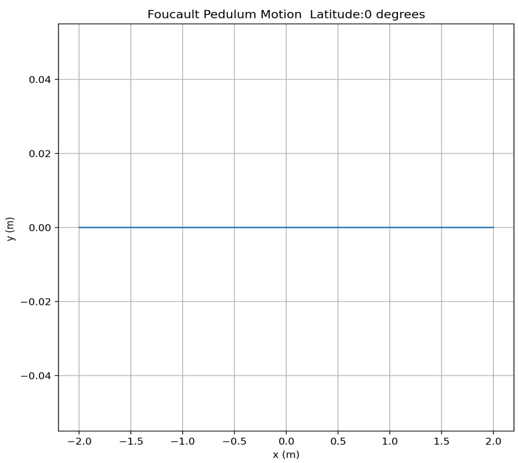


*Figure 8- latitude Equator shows no precession as there is no Coriolis force.

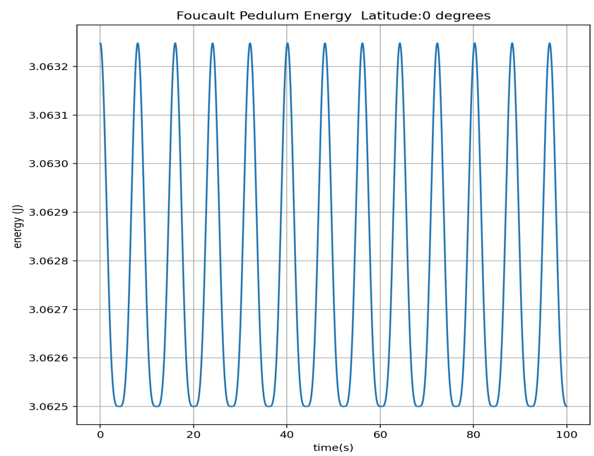


*Figure 9- Equator energy is constant as it starts and end at the same location even though there no preecession

Discussion/Conclusion

In conclusion, this study successfully models the motion of the Foucault pendulum. It displays how the precession rates do depend on latitude but also changes with it due to the Coriolis force. This demonstrates the Earth's rotation with the numerically calculated precession rates and energy conservation that are close to the theoretical expectations. At the north/south poles it will be strong and at the equator it will be weak/zero. I calculated a minimal 18.3% of error which proves the accuracy/stability. The error is slightly higher than expected but due to various numerical integration limitations this was the lowest I could obtain. The energy conservation figures were constant over time which further proves the accuracy. These findings deepen the understanding of Earth's rotational dynamics by reviewing the significance of the pendulum in the first place. It provides evidence for Earth's rotation which can be beneficial to challenge the claims regarding flat Earth theories. This experiment is one of a kind as it transformed a theoretical concept into a real world demonstration that many can still experience to this day. It ties into a larger scientific picture as the pendulum serves as a solid base for rotational dynamics as you study the motions of celestial bodies across the universe. Earth's rotation is part of the same fundamental forces and law of physics as many other formation/ evolution concepts in the universe. The Foucault pendulum acts as a reliable foundation to help uncover the mysteries of how the universe evolves over time and works. It's a vital step to understand the fundamental mechanics that govern most of the cosmos in the universe.

References

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